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PREPARED BY	APPROVED BY
RLW	RJM

SYMONS® CONE CRUSHER SOCKET DIMENSIONS AND GEAR CLEARANCES

TECHNICAL DATA
TD7-025

PROPRIETARY CODE W

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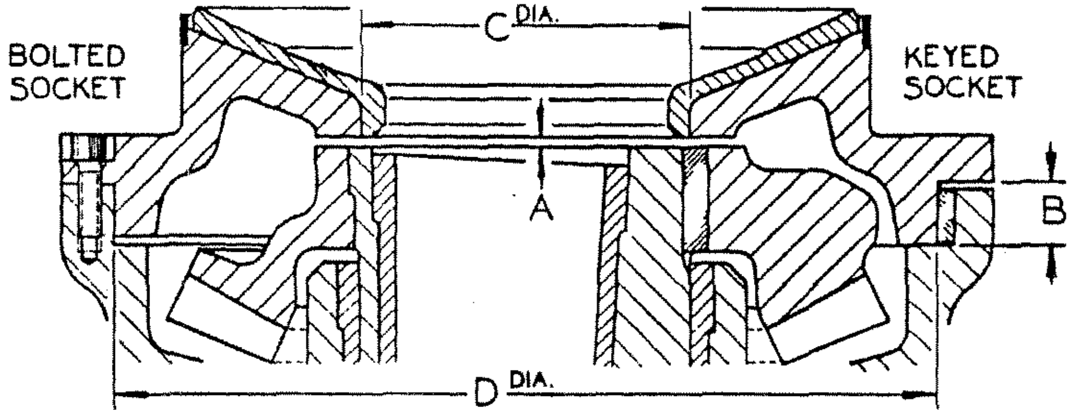

CRUSHER SIZE	TYPE SOCKET	A	B	C	D	SOCKET FIT IN MAIN FRAME
2 FT STD & SH HD	KEYED	3/16	1 33/64	8.750 / 8.752	23.005 / 23.004	.005 / .002 TIGHT
3 FT STD	KEYED & BOLTED	3/16	2 5/32	11.000 / 11.002	28.506 / 28.504	.006 / .002 TIGHT
3 FT SH HD	KEYED & BOLTED	3/16	2 5/32	12.000 / 12.002	28.506 / 28.504	.006 / .002 TIGHT
4 FT STD & 4800 STD & SH HD	KEYED	3/16	2 1/2	13.000 / 13.002	36.507 / 36.505	.007 / .003 TIGHT
4 FT STD	KEYED & BOLTED	3/16	2 1/2	14.000 / 14.002	36.507 / 36.505	.007 / .003 TIGHT
4 FT SH HD & 4 1/4 FT STD	KEYED	1/4	2 1/2	16.750 / 16.752	38.257 / 38.255	.007 / .003 TIGHT
4 FT SH HD & 4 1/4 FT STD	BOLTED	1/4	2 7/16	16.750 / 16.752	38.257 / 38.255	.007 / .003 TIGHT
5 1/2 FT STD & SH HD	KEYED	9/32	4 1/32	20.250 / 20.252	48.008 / 48.006	.008 / .003 TIGHT
5 1/2 FT STD & SH HD	BOLTED	5/16 *	3 63/64	20.250 / 20.252	48.008 / 48.006	.008 / .003 TIGHT
4 1/4 FT STD & SH HD SHORT SOCKET LINER	DOWELLED	1/4	2 5/8	24.000 / 24.002	38.259 / 38.257	.009 / .005 TIGHT

* NOTE: ON EXISTING 5 1/2 FT. BOLTED SOCKET CRUSHERS THE "A" DIMENSION IS 3/16". THIS GAP HAS BEEN INCREASED TO 5/16" TO PROVIDE ADDITIONAL GEAR CLEARANCE.

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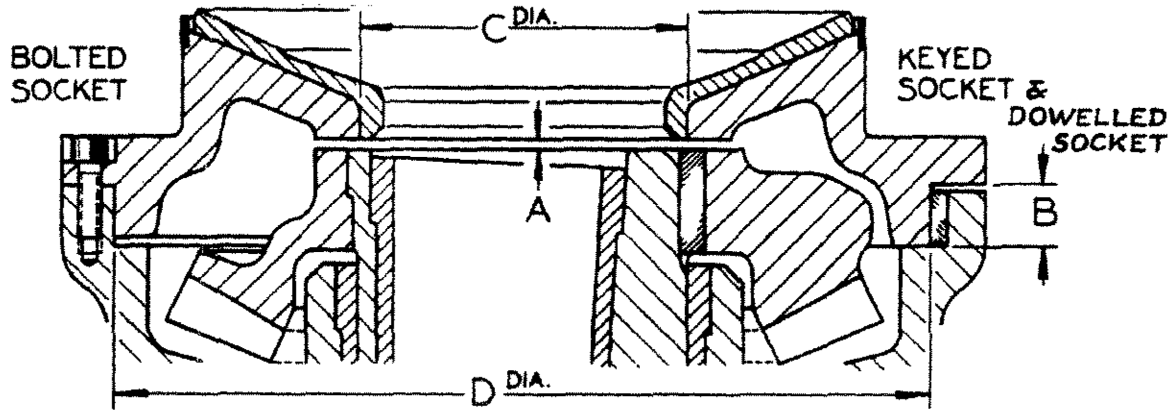

CRUSHER SIZE	TYPE SOCKET	A	B	C	D	SOCKET FIT IN MAIN FRAME
7 FT STD NORMAL DUTY	KEYED	9/32	4 1/32	24.500 / 24.503	60.509 / 60.507	.009 / .004 TIGHT
7 FT STD & SH HD	KEYED	9/32	4 1/32	25.250 / 25.252	50.509 / 60.507	.009 / .004 TIGHT
7 FT STD & SH HD	BOLTED	5/16	3 15/32	25.250 / 25.252	60.509 / 60.507	.009 / .004 TIGHT
7 FT STD & SH HD	BOLTED & DOWELLED DOWELLED	5/16	3 11/16	25.250 / 25.252	60.515 / 60.513	.015 / .010 TIGHT
7 FT STD & SH HD	KEYED, BOLTED & DOWELLED	9/32	4 1/8	25.250 / 25.252	60.509 / 60.507	.009 / .004 TIGHT
7 FT STD & SH HD SHORT SOCKET LINER	BOLTED & DOWELLED DOWELLED	5/16	3 11/16	38.006 / 38.004	60.515 / 60.513	.015 / .010 TIGHT
7 FT STD & SH HD SHORT SOCKET LINER	KEYED, BOLTED & DOWELLED	9/32	4 1/8	38.000 / 38.002	60.509 / 60.507	.009 / .004 TIGHT

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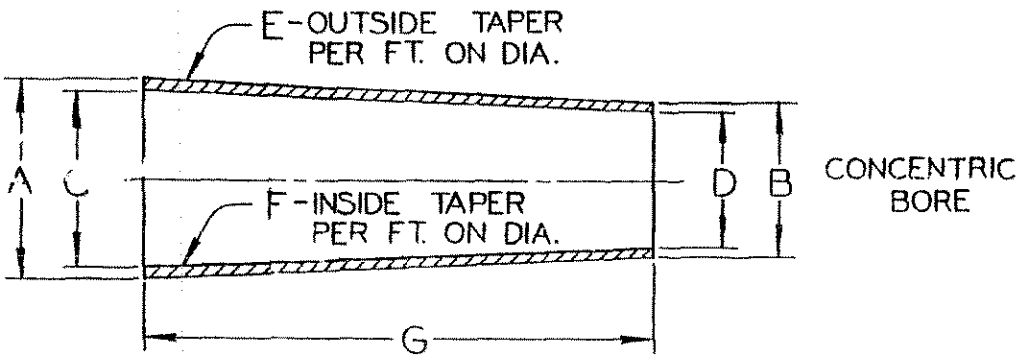
SYMONS® CONE CRUSHER
INNER ECCENTRIC BUSHING DIMENSIONS



CRUSHER SIZE AND TYPE	DRAWING NO PART CODE NO	A	B	C	D	E	F	G
2 FT STD & SH HD NORMAL DUTY	2-C-36 2206 6641	6.814 6.811	4.309 4.306	5.842 5.845	3.337 3.340	1.625	1.625	18 1/2
2 FT STD HEAVY DUTY	2-C-188 2206 6781	8.000 7.999	5.495 5.494	7.075 7.076	4.666 4.667	1.625	1.562	18 1/2
3 FT STD	3-C-49 2207 0561	8.564 8.561	6.252 6.249	7.366 7.369	5.126 5.129	1.000	.968	27 3/4
3 FT SH HD	3-C-120 2207 1401	9.377 9.374	5.908 5.905	8.242 8.245	4.773 4.776	1.500	1.500	24 3/4

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SYMONS® CONE CRUSHER INNER ECCENTRIC BUSHING DIMENSIONS

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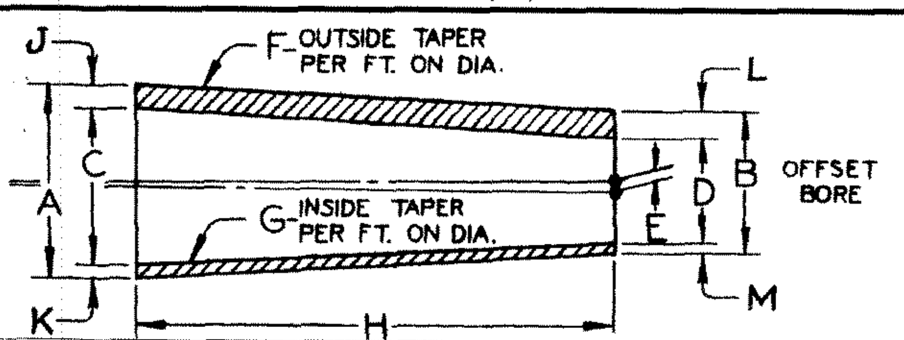
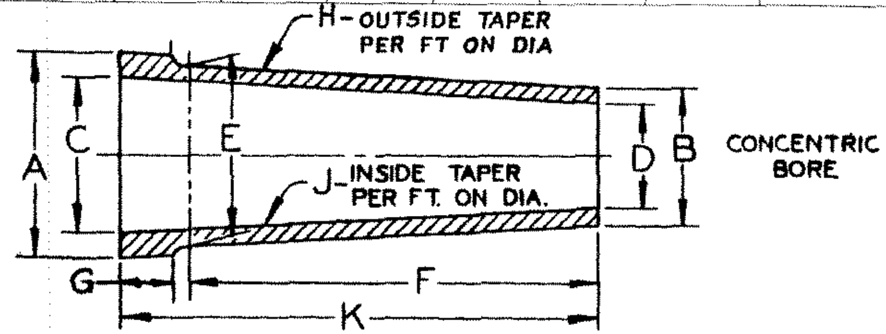
	J	K	L	M
(1)	.600	.600	.582	.528
(2)	.636	.538	.661	.563
(3)	.781	.781	.743	.654



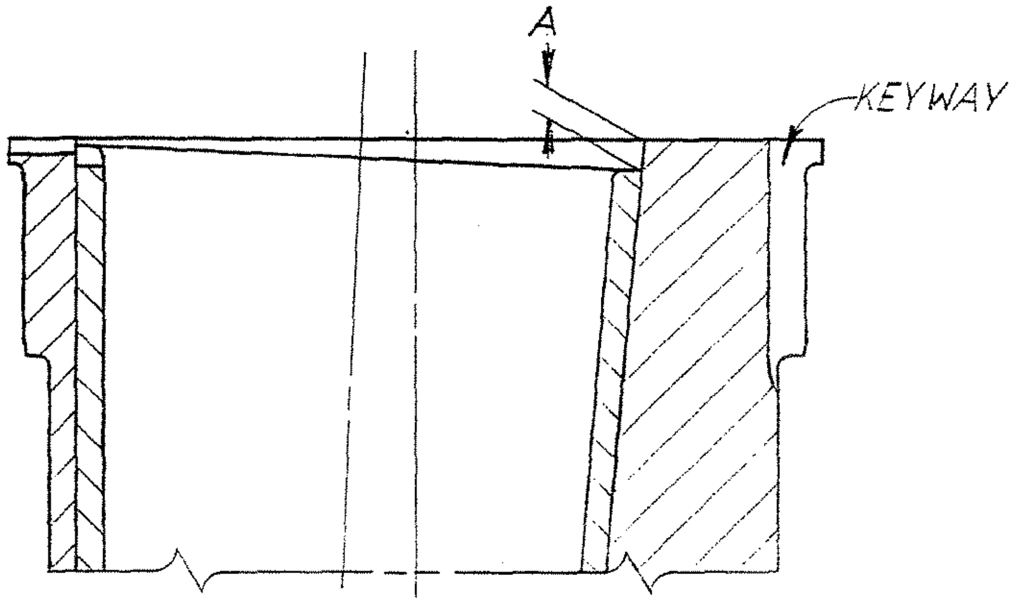
CRUSHER SIZE & TYPE	DRAWING NO PART CODE NO	A	B	C	D	E	F	G	H
4 FT STD & 4800 (1) STD & SH HD	4200-C-100 2214 4481	11.127 11.124	6.421 6.418	9.924 9.927	5.308 5.311	.027	1.625	1.593	34 3/4
5100 STD & SH HD (2) 4 1/4 FT STD & SH HD	4400-B-504 2214 3930	13.001 12.999	10.257 10.255	11.826 11.828	9.033 9.035	.049	.968	.986	34
.7 FT STD (3) NORMAL DUTY	7-C-71 2214 5741	18.313 18.311	13.500 13.498	16.748 16.750	12.101 12.103	.045	1.000	.968	57 9/16



CRUSHER SIZE & TYPE	DRAWING NO PART CODE NO	A	B	C	D	E	F	G	H	J	K
5 1/2 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	5-C-338 2214 5706	15.734 15.719	11.528 11.525	13.870 13.873	10.351 10.354	14.885	40 5/16	4 1/2	1.000	.9375	45 1/16
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	7-B-419 2214 6721	20.125 20.110	14.873 14.870	17.999 18.002	13.520 13.523	19.079	50 1/2	5	1.000	.9514	56 1/2

SYMONS® CONE CRUSHER
INNER ECCENTRIC BUSHING DIMENSIONS

LOCATION IN ECCENTRIC



MEASUREMENTS TO BE TAKEN AT KEYWAY

CRUSHER SIZE & TYPE	DRAWING NO PART CODE NO	A
2 FT STD & SH HD	3113 7351	3/8
3 FT STD	3113 8401	7/16
3 FT SH HD	3113 8751	7/16
4 FT STD & 4800 STD & SH HD	3114 0151	15/32
4 1/4 FT STD & SH HD & 5100 STD & SH HD	3114 1201	19/32
5 1/2 FT STD & SH HD	3114 2863	7/8
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	3114 4262	1-3/16
7 FT STD NORMAL DUTY	3114 3651	31/32

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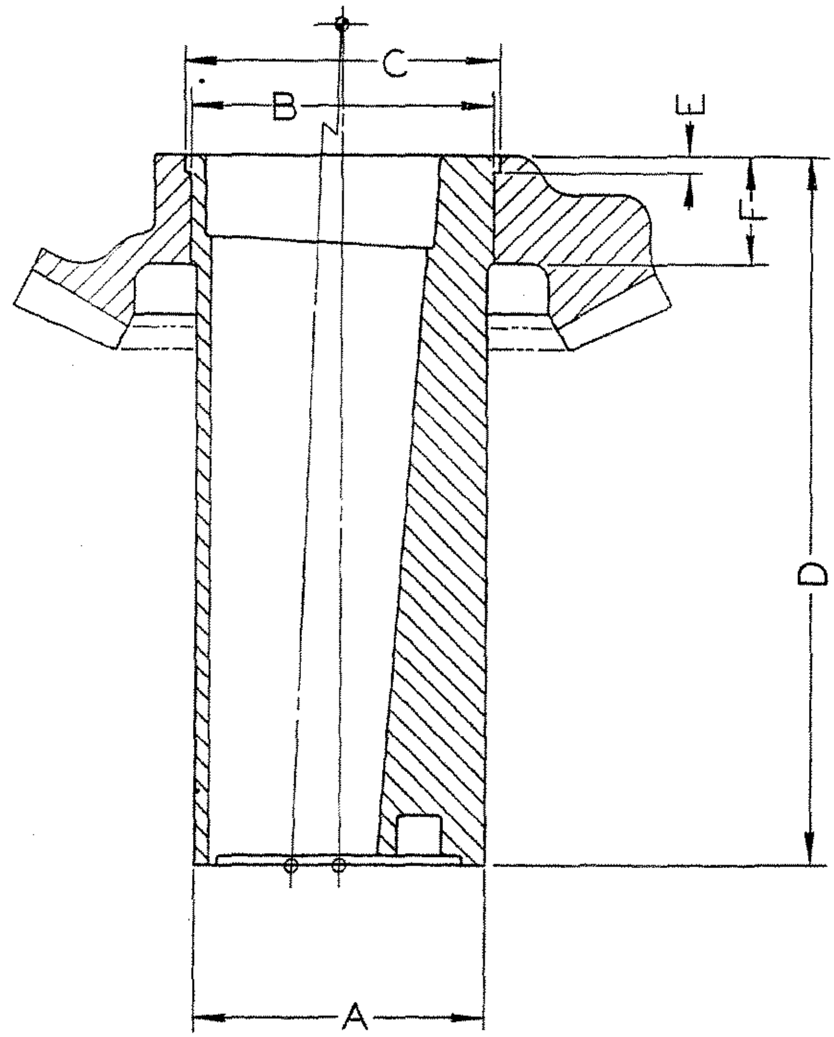
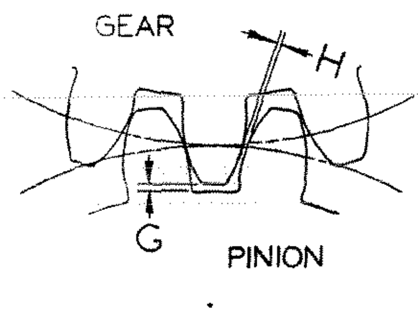
DATE OF ISSUE 12DEC62		*Metso Minerals Engineering Department*	REV A	DATE 02SEP99





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SYMONS® CONE CRUSHER ECCENTRIC DIMENSIONS WITH GEAR LOCATION AND FIT INCLUDING TOOTH BACKLASH AND CLEARANCE

Metso Minerals Engineering Department

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ECCENTRIC DIMENSIONS

CRUSHER SIZE AND TYPE	DRAWING NO PART CODE NO	A	B
2 FT STD & SH HD NORMAL DUTY	2-C-104 3113 7351	9.500 9.498	10.000 9.999
2 FT STD HEAVY DUTY	2-C-226 3113 6501	9.500 9.498	10.000 9.999
3 FT STD	3-C-116 3113 8401	11.983 11.981	12.255 12.254
3 FT SH HD	3-C-121 3113 8751	11.983 11.981	13.255 13.254
4 FT STD & 4800 STD & SH HD	4200-C-166 3114 0151	13.704 13.702	15.006 15.004
5100 & 4 1/4 FT STD & SH HD	4400-B-236 3114 1201	16.704 16.702	18.006 18.004
5 1/2 FT STD & SH HD	5-B-331 3114 2863	19.454 19.452	20.007 20.006
7 FT STD NORMAL DUTY	7-B-981 3114 3651	23.438 23.436	24.008 24.007
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY – NARROW SHOULDER	7-B-426 3114 4351	24.501 24.499	25.508 25.507
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	7-B-420 3114 4262	24.501 24.499	25.508 25.507

CRUSHER SIZE AND TYPE	C	D	E	F	ECCENTRIC SPEED-RPM
2 FT STD & SH HD NORMAL DUTY	10 15/32	19 11/16	1/2	3 3/4	357
2 FT STD HEAVY DUTY	10 15/32	19 11/16	1/2	3 3/4	393
3 FT STD	12 47/64	29 7/16	.502 .498	4	330
3 FT SH HD	13 47/64	29 7/16	.502 .498	4	320
4 FT STD & 4800 STD & SH HD	15 3/4	36	.502 .498	5	267
5100 & 4 1/4 FT STD & SH HD	18 3/4	35 7/8	1/2	5	255
5 1/2 FT STD & SH HD	20 31/32	47 1/4	3/4	7	237
7 FT STD NORMAL DUTY	24 31/32	59 1/4	3/4	8	212
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY – NARROW SHOULDER	26 15/32	59 1/4	3/4	8	212
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	26.509 26.508	59 1/4	2 31/32	8	212

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GEAR DIMENSIONS

CRUSHER SIZE AND TYPE	TOOTH COMBINATION	B	C	E	F	H TOTAL BACKLASH
2 FT STD & SH HD NORMAL & HEAVY DUTY	29 – 18	9.996 / 9.998	10 1/2	17/32	3 3/4	.020 / .040
3 FT STD	37 – 21	12.251 / 12.252	12 3/4	17/32	4	.020 / .040
3 FT SH HD	38 – 21	13.250 / 13.251	13 3/4	17/32	4	.020 / .040
4 FT STD & 4800 STD & SH HD	40 – 22	15.000 / 15.002	15 13/16	17/32	5	.020 / .040
5100 & 4 1/4 FT STD & SH HD	40 – 21	18.000 / 18.002	18 13/16	17/32	5	.030 / .050
5 1/2 FT STD & SH HD	43 – 21	20.000 / 20.001	21	25/32	7	.040 / .060
7 FT STD NORMAL DUTY	39 – 19	24.000 / 24.001	25	25/32	8	.050 / .070
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY – NARROW SHOULDER	39 – 19	25.500 / 25.501	26 1/2	25/32	8	.050 / .070
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	39 – 19	25.500 / 25.501	26.500 / 26.501	3	8	.050 / .070

CRUSHER SIZE AND TYPE	G ROOT CLEARANCE ACTUAL	G ROOT CLEARANCE MINIMUM	GEAR FIT ON ECCENTRIC	PRESS FIT IN TONS *
2 FT STD & SH HD NORMAL & HEAVY DUTY	.109	3/32	.004 / .001 TIGHT	6
3 FT STD	.109	3/32	.004 / .002 TIGHT	8
3 FT SH HD	.109	3/32	.005 / .003 TIGHT	8
4 FT STD & 4800 STD & SH HD	.125	3/32	.006 / .002 TIGHT	15
5100 & 4 1/4 FT STD & SH HD	.125	3/32	.006 / .002 TIGHT	15
5 1/2 FT STD & SH HD	.156	1/8	.007 / .005 TIGHT	20
7 FT STD NORMAL DUTY	.203	1/8	.008 / .006 TIGHT	25
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY – NARROW SHOULDER	.203	1/8	.008 / .006 TIGHT	25
7 FT STD & SH HD HEAVY & EXTRA HEAVY DUTY	.203	1/8	.008 / .006 "E" TIGHT; .009 / .007 "F" TIGHT	25

* NOTE: The preferred method of installation and removal is to "HEAT" the gear to 200°F above ambient temperature for a shrink fit.
In the event that either gear or eccentric has been metal sprayed to increase the interference fit, installation with "HEAT" is to be the ONLY method used.

When the main shaft assembly is out of the Crusher for a liner change or whenever installing the socket assembly in the Crusher, it is recommended that the proper relationship between the socket liner and inner eccentric bushing be checked. At the same time it is advisable to check the head ball for wear in respect to the tapered portion of the main shaft. A pair of sheet metal templates obtained from the factory, are needed to make these checks. One template is to check the condition of the socket liner and the other is to check the condition of the head ball.

When the templates are used, a check is being made of the relationship between socket liner, inner eccentric bushing, head ball and main shaft. This relationship is known as the geometry of the Crusher – when correct, the Crusher is a smooth running machine capable of transmitting the greatest crushing force possible with the least energy output. When the geometry is incorrect, the Crusher loses bearing clearance and generates heat.

The socket liner which provides the bearing surface for the head is machined so that its outer edge is normally the only portion which will be in contact with the head initially. As the spherical bearing surface of the socket liner wears, the main shaft assembly drops further into the inner eccentric bushing causing the running clearance between the main shaft and inner eccentric bushing to be reduced. When this wear occurs, it may cause the head to revolve or spin at an abnormally fast rate when the Crusher is running empty. (0 to 10 RPM is considered normal) In conjunction with the spinning head, there will be a possible loss of oil and an increase in oil temperature.

When a spinning head condition is noticed, the socket liner most likely will have to be replaced unless the socket can be shimmed to compensate for the socket liner wear. In order to determine if the socket liner needs replacing or the socket can be shimmed, proceed as follows:

1) Remove the main shaft assembly from the Crusher.
2) Place the socket liner template on the socket liner so that it is positioned over the low side of the inner eccentric bushing which is next to the eccentric keyway. Make sure that the template contacts the inner diameter of the socket liner and the spherical surface of the socket liner as

shown in the illustration SOCKET LINER CHECKING TEMPLATE.

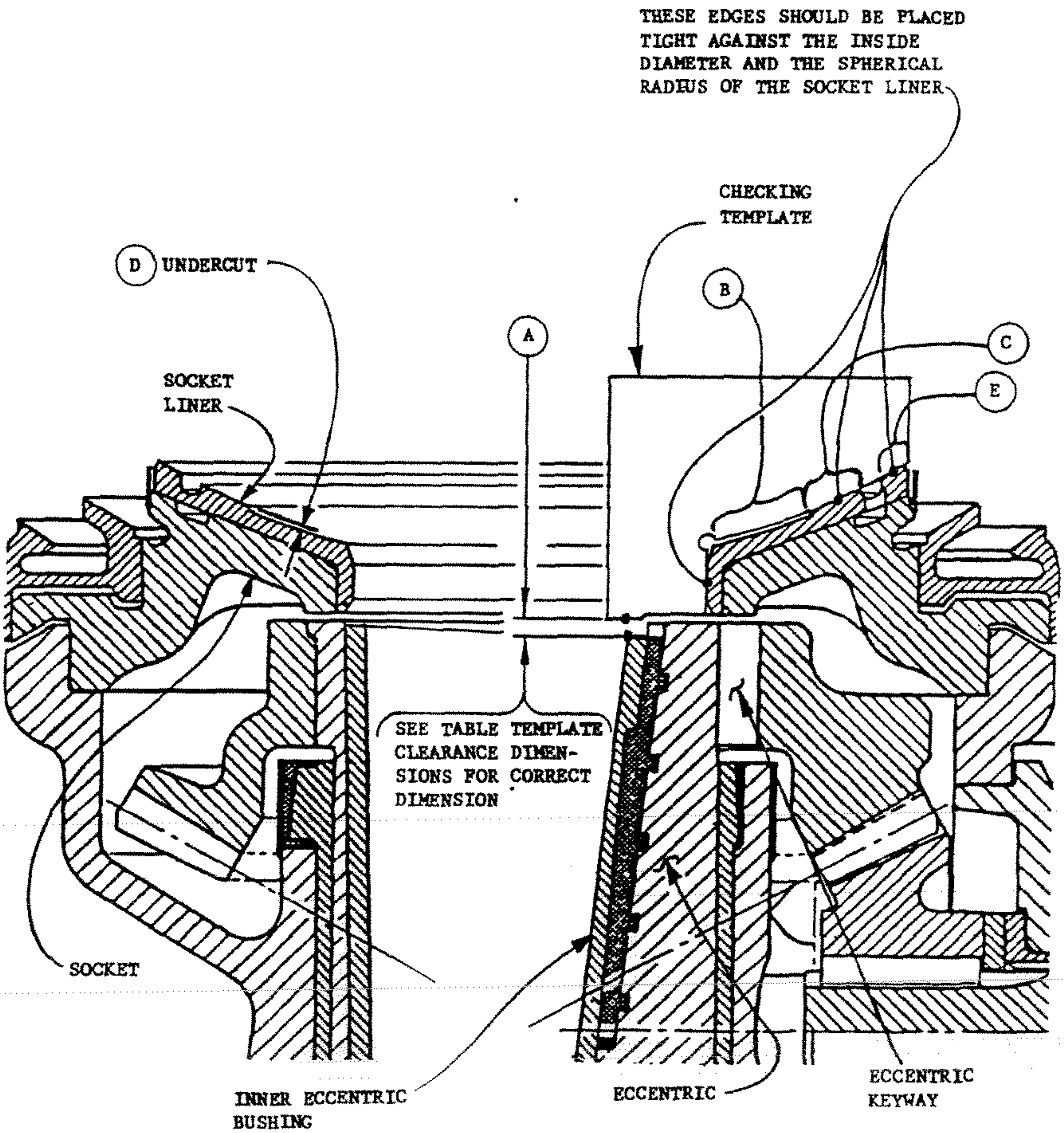
3) Take a measurement at "A". This dimension should be as shown in the table TEMPLATE CLEARANCE DIMENSIONS. RECORD THE "A" DIMENSION. There must be a gap at "B", however, it is not necessary to measure this clearance at this time.

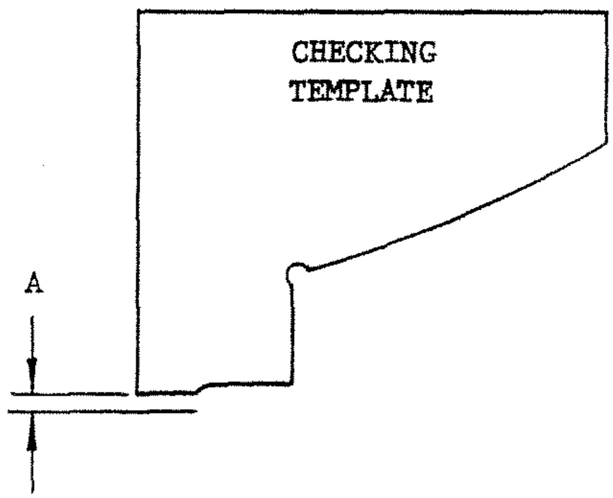
In the event the measurement shows that the "A" dimension is less than it should be, SOME corrections can be made by shimming the socket. Check to see if the socket had been shimmed previously. This can be determined by checking the clearance between the socket flange and the main frame as shown in the illustrations SOCKET SEATING SURFACES or the shim can be seen between the main frame and the outside flange of the socket. If there is a shim it may have to be removed so that the correct "A" dimension can be obtained.

4) At this time check to see where the template makes contact with the spherical bearing surface on the socket liner, if there is a gap at the outer surface at "E" and contact is made only on the inner surface at "C", chances are that the Crusher will throw oil when it is reassembled. This can be corrected by scraping or remachining the socket liner so that the contact is made on the outer portion, at "E". Scrape only the lower two-thirds of the socket liner at "C". Scrape enough metal from the surface of the socket liner so that the wear pattern will again be at the outside edge of the liner at "E".

If oil is being thrown out of the head sealing arrangement after the installation of the new socket liner, the ball of the head has worn to a different radius and does not mate with the spherical surface of the new socket liner. Throwing oil means the worn head is bearing on the inside portion of the new liner leaving the outside portion of the liner and head ball open. See the illustration WORN HEAD WITH NEW SOCKET LINER. This opening, even though only a few thousandths of an inch, will cause oil to be thrown out over the sealing ring and eventually out of the Crusher around the socket area.

SOCKET LINER
CHECKING TEMPLATE

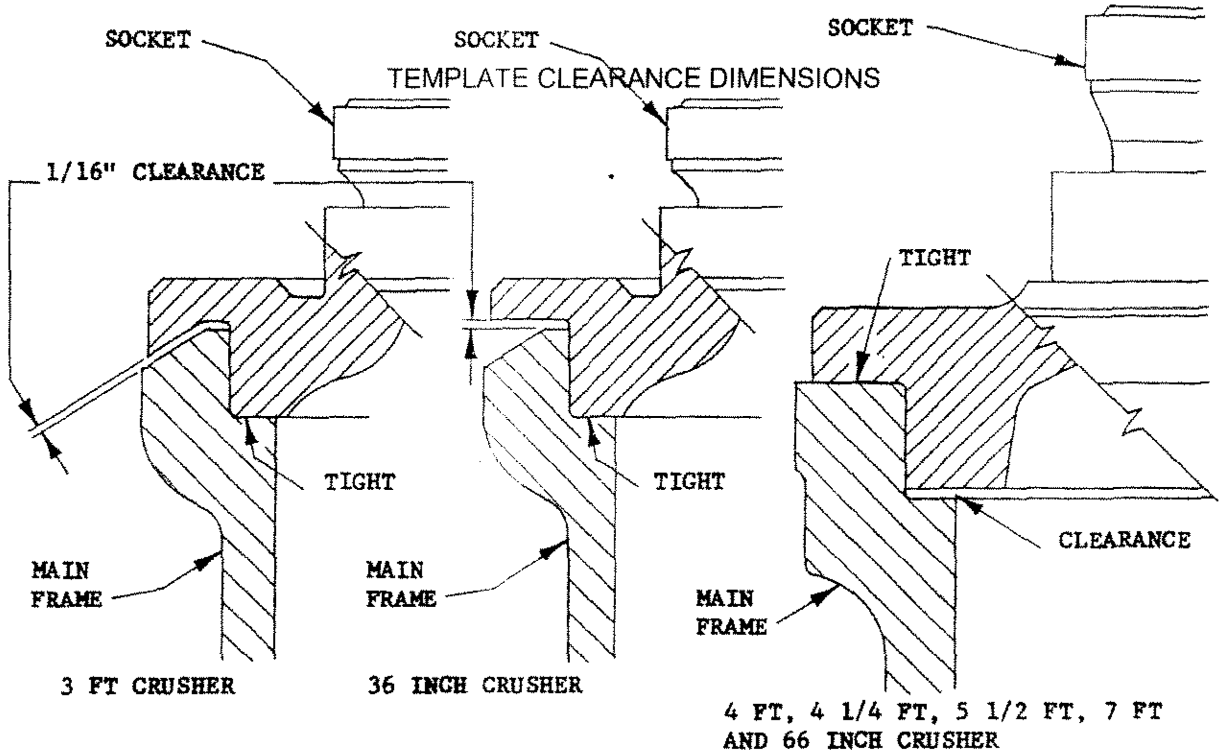




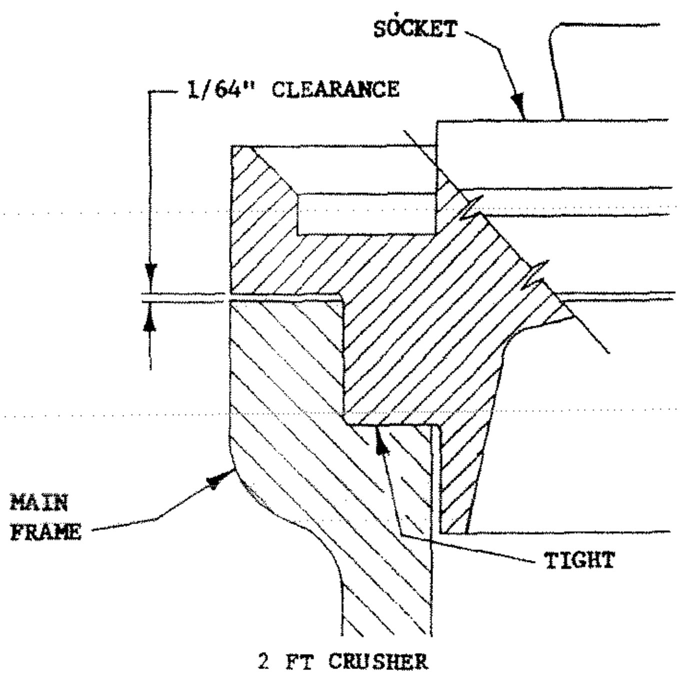
See Illustration—
SOCKET LINER
CHECKING TEMPLATE
for use of template

CRUSHER SIZE	ACTUAL CLEARANCE	MINIMUM CLEARANCE	MAXIMUM CLEARANCE
2 Ft., 3 Ft. and 36 In.	1/2	7/16	5/8
4 Ft., 4-1/4 Ft., 5-1/2 Ft., 7 Ft., 48 In., 66 In. and 84 In.	1/2	13/32	11/16
All Dimensions are in Inches			

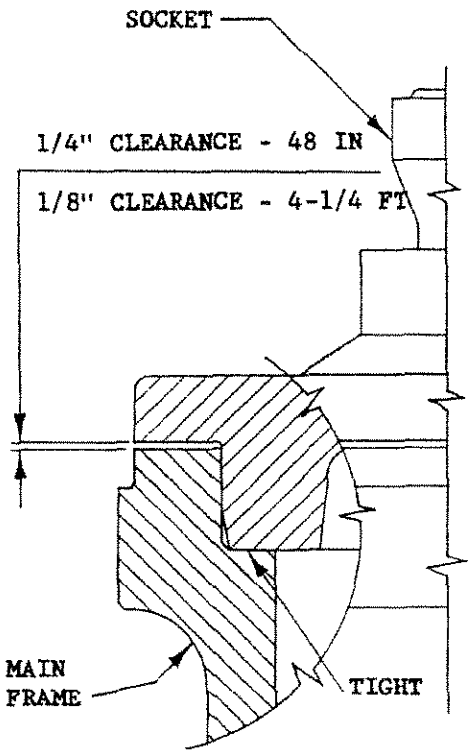
TEMPLATE CLEARANCE DIMENSIONS



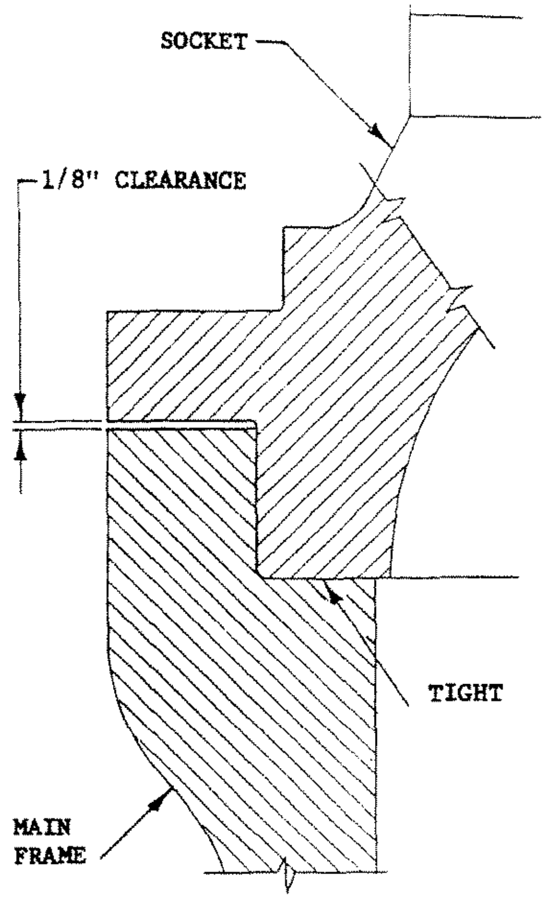
BOLTED SOCKET SEATING SURFACES



KEYED SOCKET SEATING SURFACE



48 INCH & 4-1/4 FT CRUSHER



84 INCH & 7 FT CRUSHER

DOWELLED SOCKET SEATING SURFACES

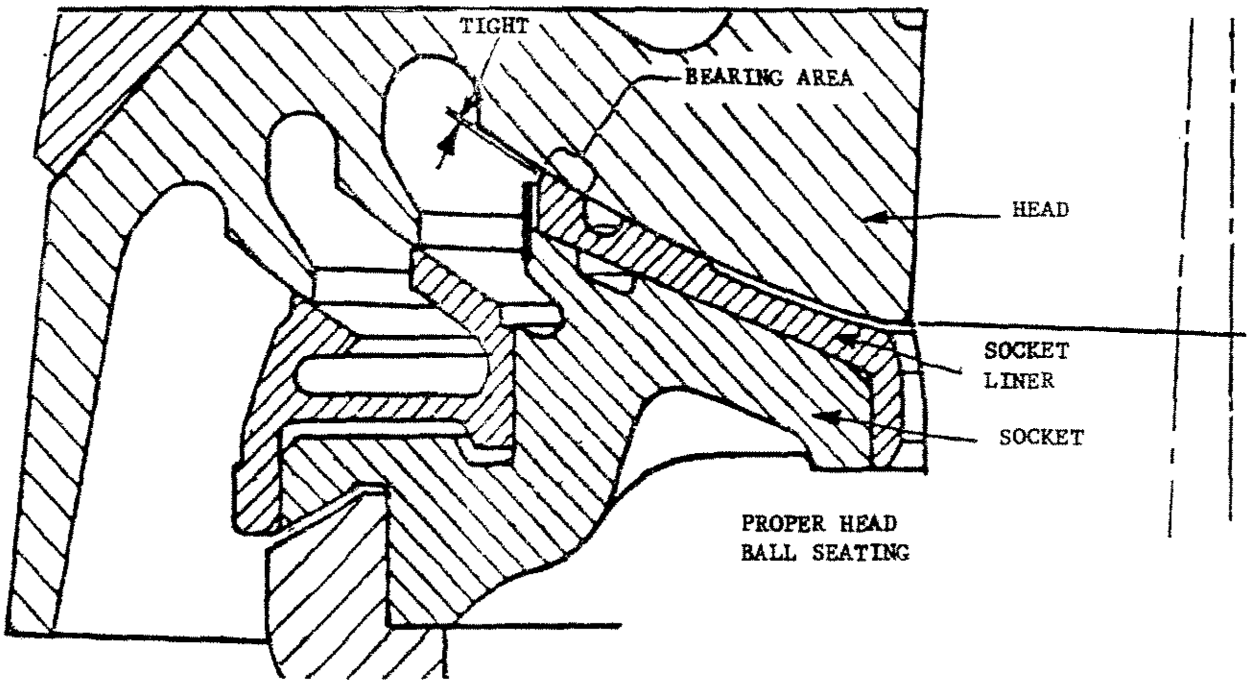
Another method of checking the contact pattern is to put machinist's bluing on the socket liner bearing surface. Then install the main shaft assembly into the Crusher and run it for a few minutes. Remove the main shaft assembly and inspect the wiping pattern of the head on the liner. The bluing will wear off the liner where it made contact with the head ball and still be on in the area where no contact is made. This will show at a glance if and where the socket liner should be re-worked. The correct pattern should show that the head is seating evenly around the outside edge of the socket liner at "E".

THE SOCKET LINER MUST BE REPLACED WHEN SURFACE "C" HAS WORN TO THE EXTENT OF BLENDING INTO THE UNDERCUT SURFACE AT "B".

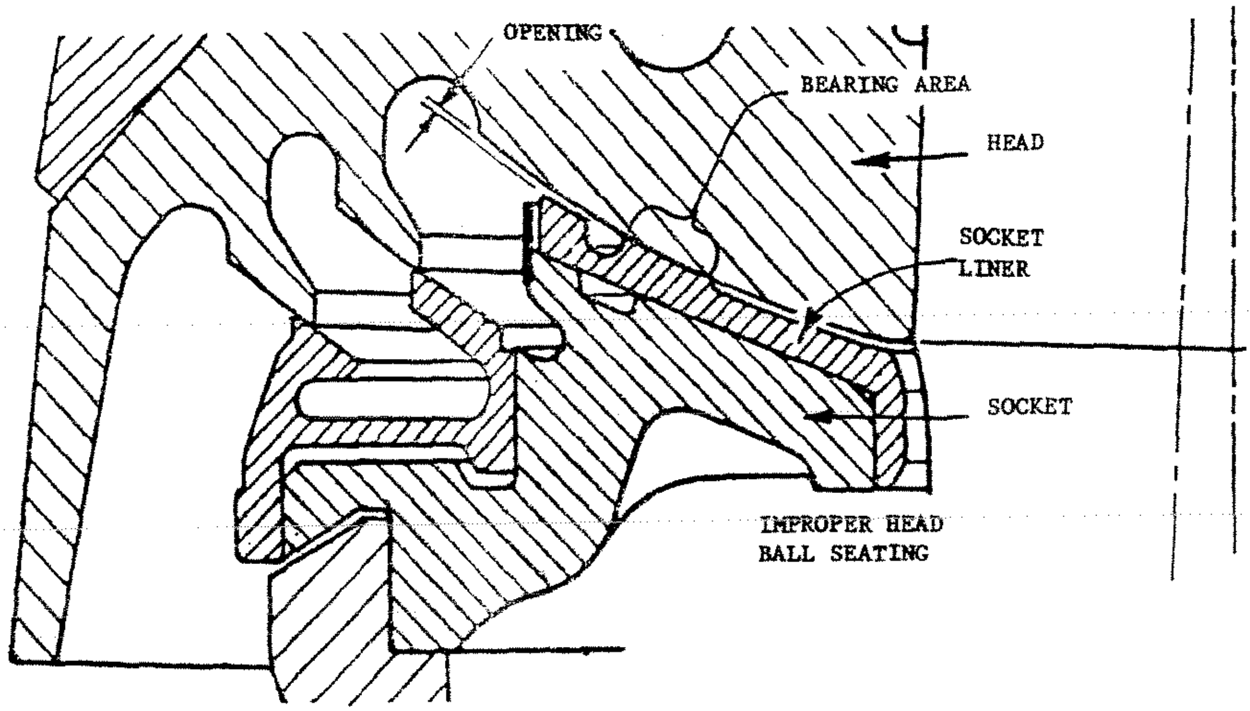
5) At this time, the wear on the head ball should also be measured, as the correct relationship between the main shaft assembly and eccentric assembly depends on both the socket liner location in regard to the inner eccentric bushing AND the amount of wear on the head.

6) Place the main shaft and head template so that the hooked portion on the bottom of the template seats firmly on the bottom of the main shaft. The lower, middle and top contact points of the template should be tight against the tapered portion of the shaft as shown in the illustration MAIN SHAFT AND HEAD CHECKING TEMPLATE. With the template held in this position, the amount of wear on the head ball can be easily seen.

7) Make four measurements, every 90°, of the clearance (if any) between the template and the head ball. Take an average of these four dimensions and RECORD THIS DIMENSION. In the event the measurements show the wear is less than 1/4", SOME corrections can be made by shimming the socket. If the wear on the head is between 1/4" and 3/8", it is NOT recommended to shim up the socket to compensate for this amount of wear; the head ball should be built up with weld and remachined.



PROPER HEAD BALL SEATING



IMPROPER HEAD BALL SEATING

WORN HEAD WITH NEW SOCKET LINER

8) If the Crusher was losing oil over the socket liner, make two additional measurements; one at "F", the other at "G". If the dimension at "F" is LESS than that at "G", the head can be used as is. Four measurements, every 90°, should be taken. Should the measurement at "F" be MORE than at "G", the socket liner will have to be scraped or remachined to obtain the proper bearing contact with the head ball or the head ball may have to be remachined and polished to restore the head to the original contour. The head must make contact at the outer surface of the socket liner.

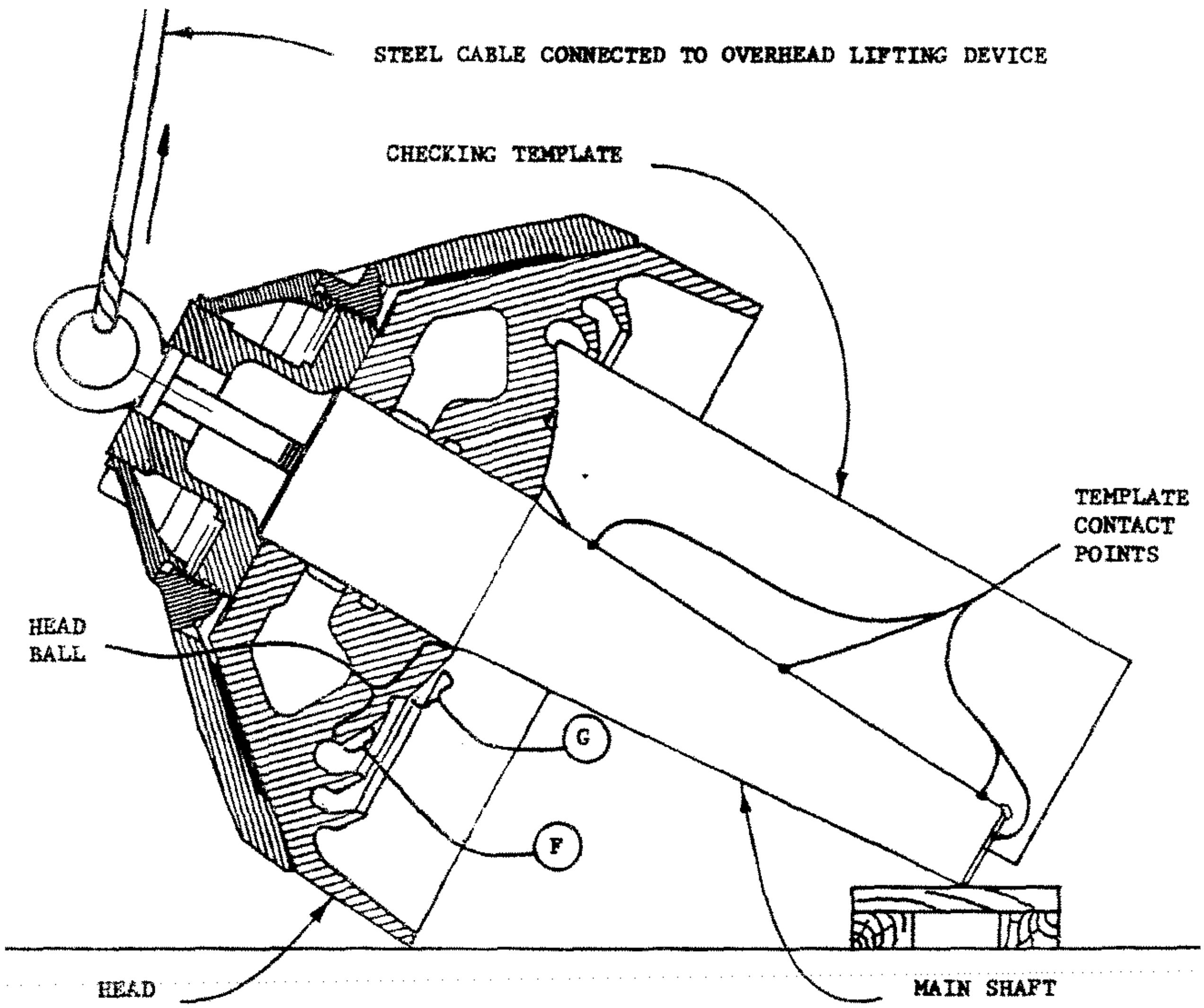
9) It is most important to know how to properly use BOTH templates. The procedure recommended for obtaining the correct dimension at "A" is as follows:

   A. With the socket assembly installed in the Crusher, measure the dimension at "A" using the socket liner checking template. See Step 2.

   B. Check the wear on the spherical bearing surface of the socket liner using the same template. See Step 4.

   C. Measure the wear on the head ball using the main shaft and head checking template. See Steps 6 and 8.

Dimension "A" must be obtained by a combination of checks. If the socket liner and head ball are within the wear limitations as previously described, the correct Dimension "A" as shown in table TEMPLATE CLEARANCE DIMENSIONS can be obtained by shimming the socket. DO NOT SHIM THE SOCKET LINER.



MAIN SHAFT AND HEAD CHECKING TEMPLATE

EXAMPLE:

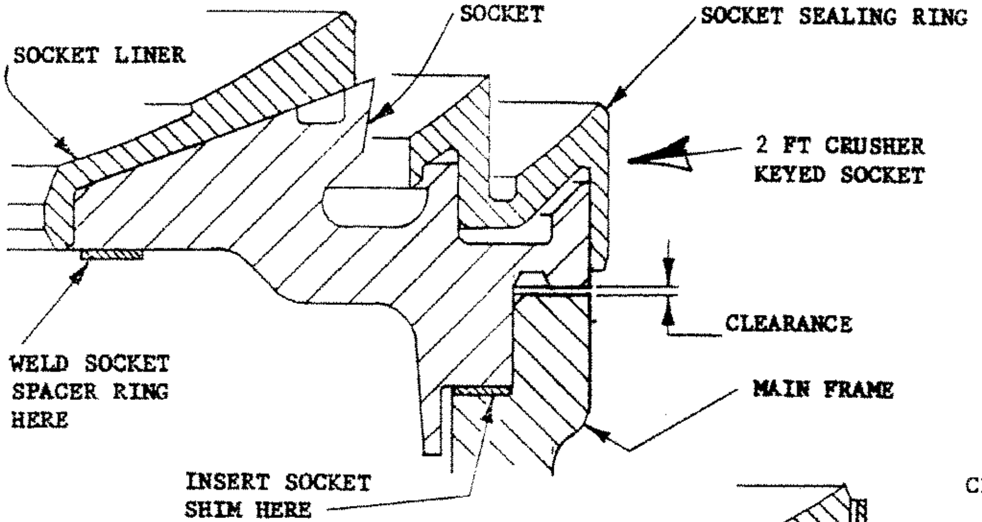
A. Measured dimension at "A" shows 3/8". Also check surface "C" on the socket liner relative to the undercut at "B" to see if the socket liner is serviceable.

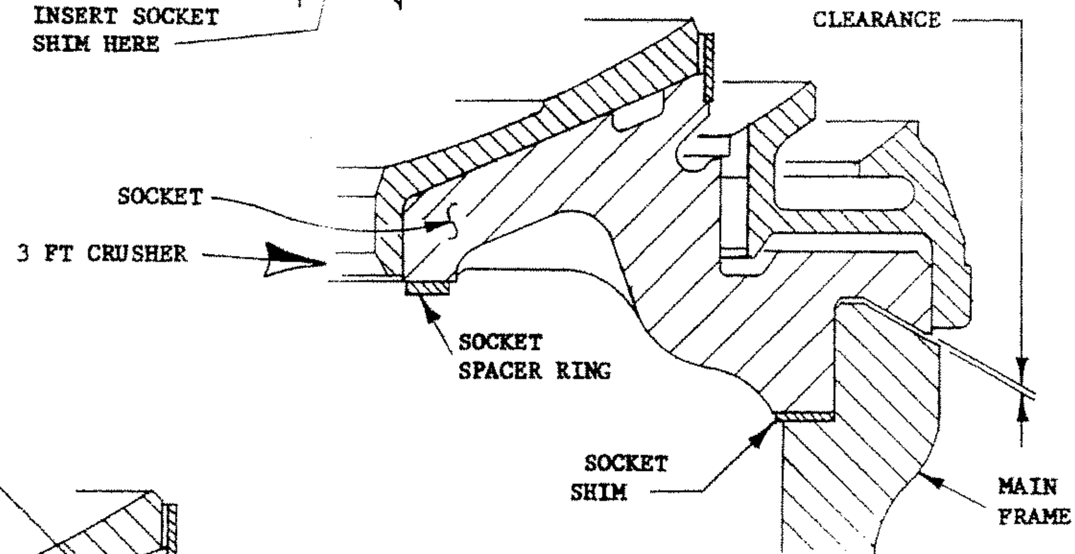
B. A check of the head ball shows 1/8" wear.

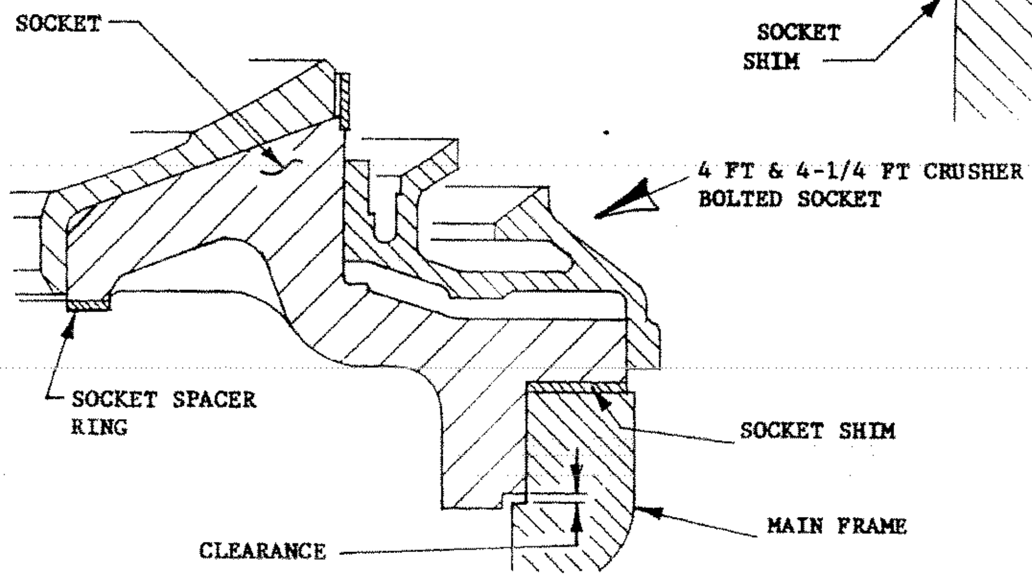
C. This combination of checks would indicate that the position of the main shaft relative to the inner eccentric bushing is INCORRECT. The main shaft assembly is 1/4" too low. To correct this situation, shimming of the socket is all that is required. A shim under the socket seating surface, in this example 1/4" thick, should be installed as shown in the illustration INSTALLING SHIMS. Dimension for the socket shim are shown in the table SOCKET SHIM DIMENSIONS.

10) Whenever the socket is shimmed beneath the socket seating surface, a spacer ring equal in thickness to that of the socket shim MUST be welded to the underside of the socket at "H" as shown in the illustration INSTALLING SHIMS. Dimensions for the socket spacer rings are shown in the table SOCKET SPACER RING DIMENSIONS. The purpose for the socket spacer ring at "H" is to maintain the clearance between the underside of the socket and the top of the eccentric and gear as shown in the table SOCKET AND ECCENTRIC CLEARANCES. If this clearance exceeds the MAXIMUM CLEARANCE shown in the table, the eccentric assembly could lift off the upper step bearing plate when the oil is cold or when tramp iron passes through the Crusher or when the adjustment ring jumps off its main frame seating surface.

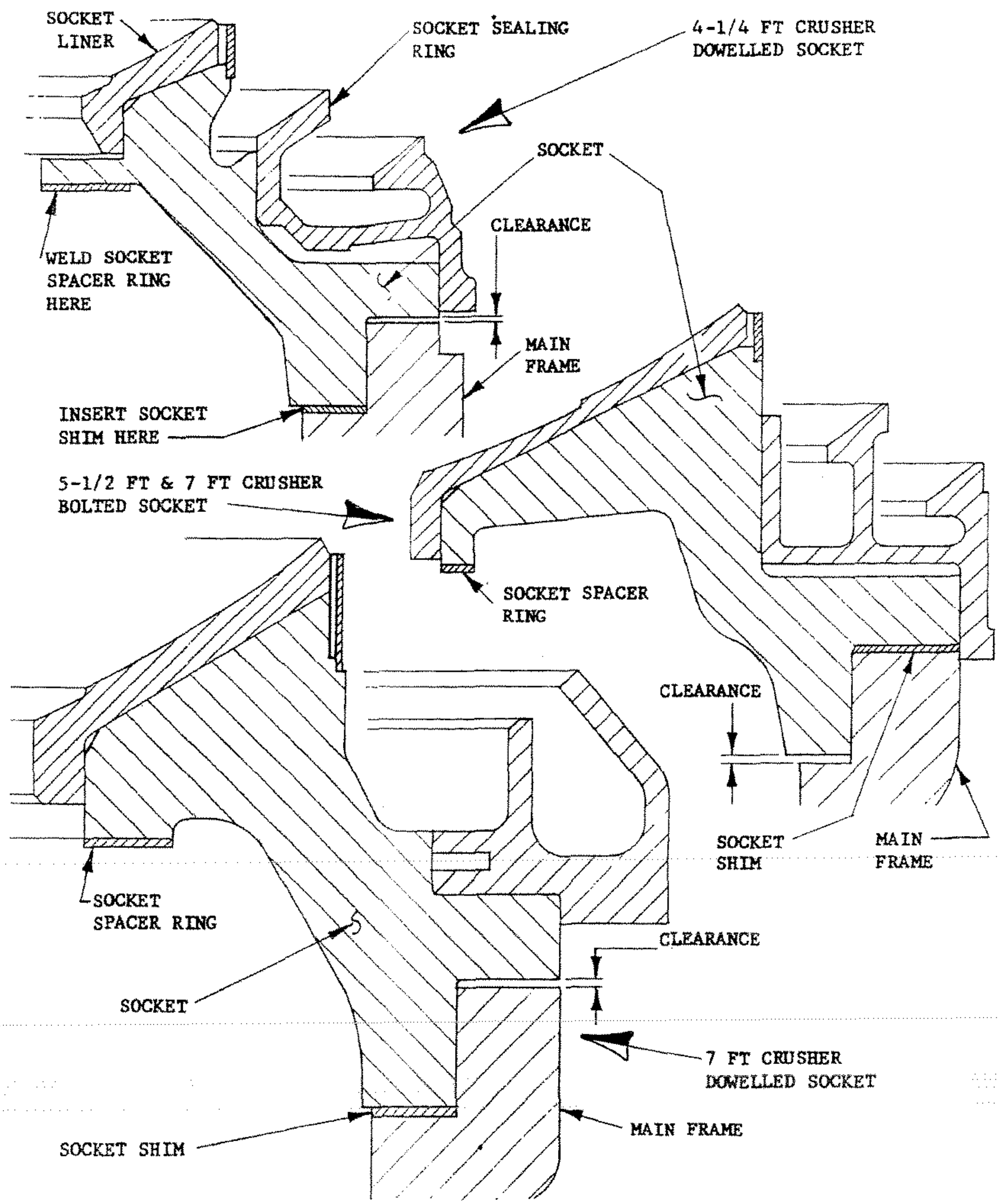
Should the eccentric assembly lift off the upper step bearing plate, it MAY NOT come back down on the plate properly and serious damage to the Crusher could result.



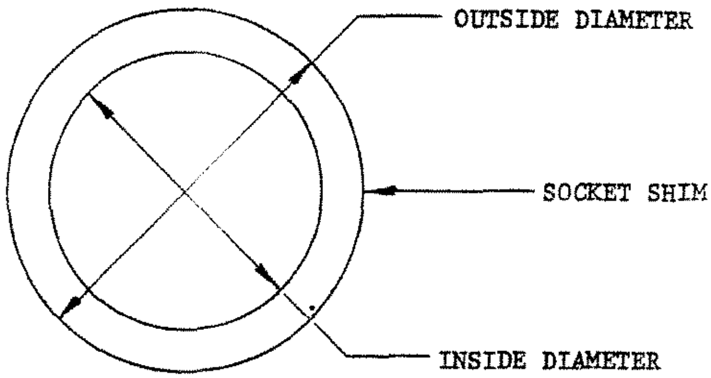




INSTALLING SHIMS



INSTALLING SHIMS



CRUSHER SIZE	OUTSIDE DIAMETER	INSIDE DIAMETER	THICKNESS
2 Ft.	22 15/16	21	To suit the wear condition shown by the socket liner checking template
3 Ft. and 36 In.	28 1/4	26 1/2	
4 Ft.	41 1/2	36 5/8	
4-1/4 Ft. Bolted Socket	42 1/2	38 1/2	
4-1/4 Ft. Dowelled Socket and 48 In.	38 3/16	36	
5-1/2 Ft. and 66 In.	55	48 1/4	
7 Ft. Bolted Socket	66 3/4	60 3/4	
7 Ft. Dowelled Socket And 84 In.	60 3/8	55 7/16	
All Dimensions are in Inches			

SOCKET SHIM DIMENSIONS

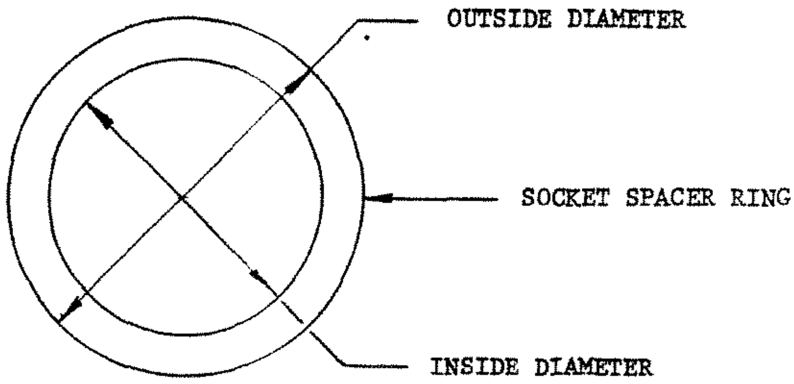
The socket assembly must be removed from the Crusher whenever shimming the socket to permit the installation of the shim under the socket seating surface and the welding of the socket spacer ring at "H". If the shim and the spacer ring are made from segments (more than one piece) instead of one piece, they should be made so that the segments when placed in position make a complete ring with no gaps at the joints. If, where the socket shim is to be placed there are tapped or drilled holes in either the main frame or socket, the shim must be made to provide clearance for these holes.

The socket spacer ring should be welded to the underside of the socket at "H" so that the ring is as flat as possible. All welds are to be grounded flush with the flat surface of the spacer ring.

THE PROCEDURE DESCRIBED IN STEP 10 DOES NOT APPLY TO THE 36 IN., 48 IN., 66 IN. AND 84 IN. GYRADISC CRUSHERS. THE UNDERSIDE OF THE SOCKETS ON THESE MACHINES DOES NOT PERMIT THE ADDITION OF THE SOCKET SPACER RING AT "H".

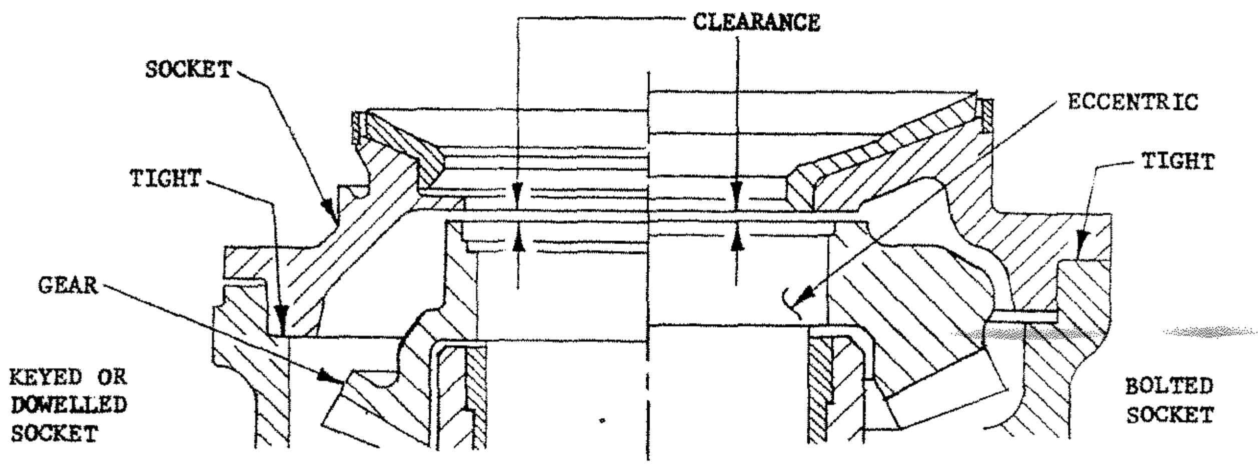
ON THE 36 IN., 48 IN. AND 84 IN. SIZES THERE IS NO NEED TO ADD THE SOCKET SPACER RING AT "H" AS THE ECCENTRIC ASSEMBLY CANNOT LIFT OFF THE UPPER STEP BEARING PLATE BECAUSE OF THE DEEP SHOULDER ON THE PLATE.

ON THE 66 IN. SIZE THE SOCKET SPACER RING MUST BE BRAZED TO THE TOP OF THE CAST IRON COUNTERWEIGHT AS SHOWN IN THE ILLUSTRATION COUNTERWEIGHT SPACER RING. BUT SINCE THE PORTION OF THE COUNTERWEIGHT THAT IS NEAR THE UNDERSIDE OF THE SOCKET LINER DOES NOT MAKE A FULL RING, THE SPACER RING MUST BE MADE TO CONFORM TO THE COUNTERWEIGHT PROFILE, APPROXIMATELY 180°.



CRUSHER SIZE	OUTSIDE DIAMETER	INSIDE DIAMETER	THICKNESS
2 Ft.	12	9 1/4	Equal In Thickness To The Socket Shim
3 Ft. Std.	12 1/2	11 1/2	
3 Ft. Sh. Hd.	13 1/2	12 1/2	
4 Ft.	15 1/2	14 1/2	
4-1/4 Ft. Bolted Socket	18	17 1/4	
4-1/4 Ft. Dowelled Socket	24 (4 pieces	19 3/4 required)	
5-1/2 Ft.	22	20 3/4	
7 Ft. Bolted Socket	32	25 3/4	
7 Ft. Dowelled Socket	42 3/4	38 1/2	
All Dimensions are in Inches			

SOCKET SPACER RING DIMENSIONS



CRUSHER SIZE	ACTUAL CLEARANCE	MINIMUM CLEARANCE	MAXIMUM CLEARANCE
2 Ft.	3/16	3/32	1/4
3 Ft.	3/16	3/32	1/4
4 Ft.	3/16	3/32	1/4
4 1/4 Ft.	1/4	1/8	5/16
5 1/2 Ft.	5/16	3/16	3/8
7 Ft.	5/16	3/16	3/8

All Dimensions are in Inches

SOCKET AND ECCENTRIC CLEARANCES

11) After the shim and spacer ring have been installed in the main frame and to the socket, the socket is to be re-installed in the Crusher as described in INSTALLING THE SOCKET in the instruction manual, Section 6.

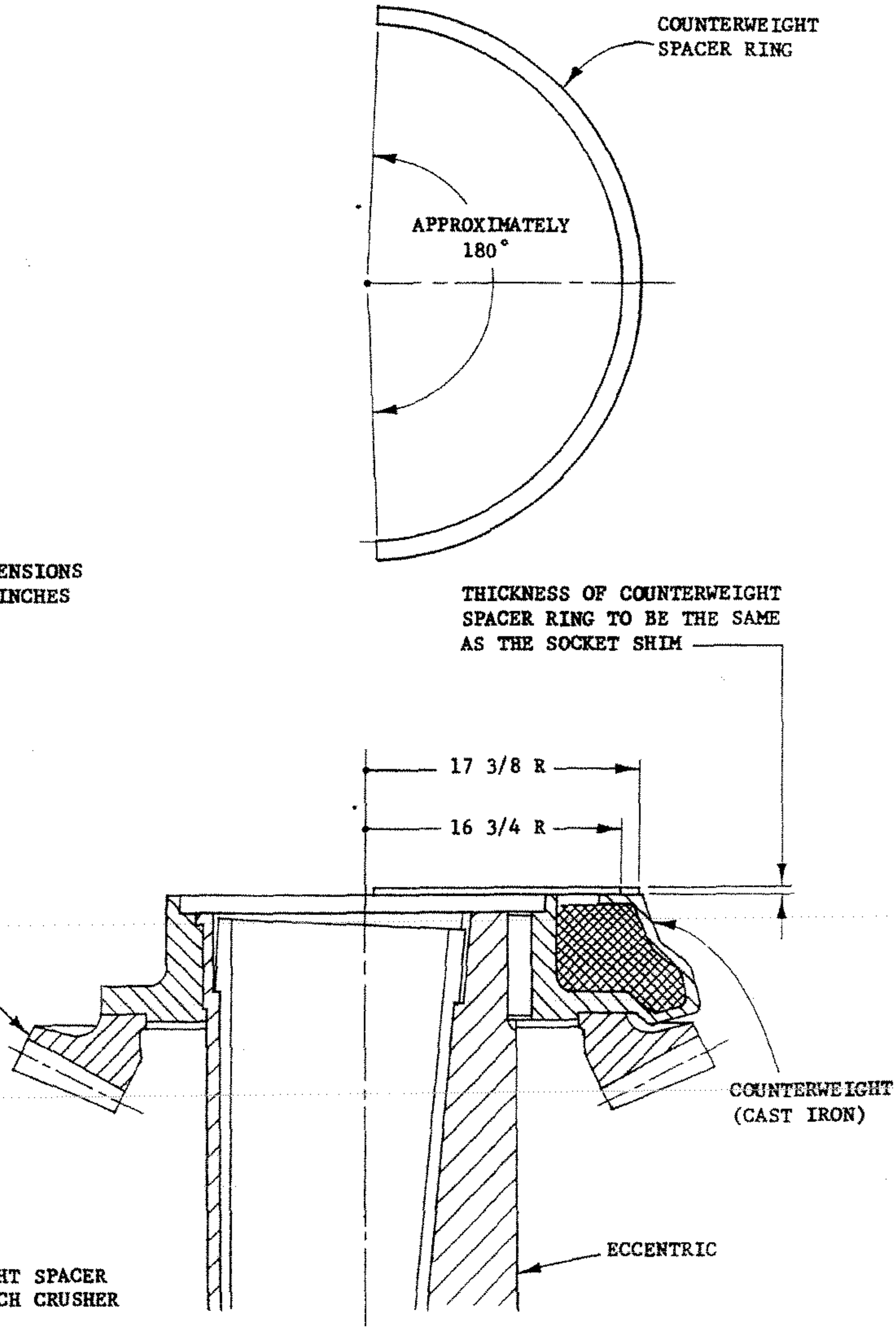
12) Recheck the 1/2" gap at "A" using the socket liner checking template after the socket is installed to be sure that the proper thickness of shim has been used.

Whenever replacing or reconditioning any of the following parts: socket liner, head ball, inner eccentric bushing or using a different main shaft assembly, it may be necessary to remove the socket shim AND the socket spacer ring; consequently, the entire checking procedure should be repeated.

If the head ball has almost reached its maximum wear limitation of 1/4", and the socket liner has almost reached its wear limitation; that is, no undercut, carefully read the following instructions.

If a Cone Crusher is equipped with a standard or air sealing arrangement, the springs under the socket sealing ring may be compressed too much and actually hold the head off the surface of the socket liner and cause the Crusher to lose or throw an excessive amount of oil. If any interference or excessive spring pressure is encountered, replace the socket liner and/or restore the head ball to its original contour. THIS CONDITION CANNOT BE CORRECTED BY SHIMMING.

If a Gyradisc Crusher is equipped with a grease or air sealing arrangement, the sealing ring may be making contact with the top of the socket. This condition will actually hold the head off the surface of the socket liner and cause the Crusher to lose or throw an excessive amount of oil. If any interference between sealing ring and socket is encountered, replace the socket liner and/or restore the head ball to its original contour. THIS CONDITION CANNOT BE CORRECTED BY SHIMMING.



ALL DIMENSIONS ARE IN INCHES

COUNTERWEIGHT SPACER RING - 66 INCH CRUSHER

If a Cone or Gyradisc Crusher is equipped with a water sealing arrangement, check to see that there is clearance between the baffle ring welded to the head and the water chamber bolted to the socket. If any interference is encountered, replace the socket liner and/or restore the head ball to its original contour. THIS CONDITION CANNOT BE CORRECTED BY SHIMMING.

metso minerals

PREPARED BY	APPROVED BY
RLW	RJM

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SYMONS® CONE & NORDBERG GYRADISC® CRUSHERS INSTRUCTIONS FOR MAINTENANCE OF PROPER MAIN SHAFT, SOCKET LINER AND INNER ECCENTRIC BUSHING GEOMETRY USE OF CHECKING TEMPLATES

Metso Minerals Engineering Department

TEMPLATE PART CODE NUMBERS

SOCKET LINER CHECKING TEMPLATES

CRUSHER SIZE	PART CODE	CRUSHER SIZE	PART CODE
3 Ft. Std.	7962 5603	36 In. w/o Socket Liner Relief	7962 7001 (Old Style)
3 Ft. Sh. Hd.	7962 5605	36 In. w/ Socket Liner Relief	7962 7002
4 Ft.	7962 5702	48 In. w/o Socket Liner Relief	7962 7201 (Old Style)
4-1/4 Ft. Bolt Socket	7962 5802	48 In. w/ Socket Liner Relief	7962 7202
4-1/4 Ft. Dowelled Socket	7962 5808	66 In. w/o Socket Liner Relief	7962 7401 (Old Style)
5-1/2 Ft.	7962 5903	66 In. w/ Socket Liner Relief	7962 7402
7 Ft. Normal Duty	7962 6002	84 In. w/o Socket Liner Relief	7962 7500 (Old Style)
7 Ft. Heavy and Extra Heavy Duty-Long Socket Liner	7962 6007	84 In. w/ Socket Liner Relief	7962 7502
7 Ft. Heavy and Extra heavy Duty-Short Socket Liner	7962 6006		

MAIN SHAFT AND HEAD CHECKING TEMPLATES

CRUSHER SIZE	PART CODE	CRUSHER SIZE	PART CODE
3 Ft. Std.	7962 5602	36 In.	7962 7000
3 Ft. Sh. Hd.	7962 5604	48 In.	7962 7200
4 Ft.	7962 5701	66 In.	7962 7400
4-1/4 Ft.	7962 5801	84 In.	7962 7501
5-1/2 Ft.	7962 5902		
7 Ft. Normal Duty	7962 6001		
7 Ft. Heavy and Extra Heavy Duty	7962 6005		